

Faculty of Science, Engineering and Built Environment

SEE212 Power Electronics

Deakin University Unit Guide

Trimester 2, 2025

CONTENTS

Welcome 2
Who is the unit team? 2
Unit chair: leads the teaching team and is responsible for overall delivery of this unit 2
Other members of the team and how to contact them 2
Administrative queries 2
About this unit 2
Unit development in response to student feedback 2
Learning Outcomes 3
Assessing your achievement of the unit learning outcomes 3
Summative assessment task 1 3
Summative assessment task 2 4
Your learning experiences in this unit 5
Educator-facilitated (scheduled) learning activities - on-campus unit enrolment 5
Educator-facilitated (scheduled) learning activities - online unit enrolment 5
Typical study commitment 5
Note 6
Unit learning resources 6
Essential learning resources 6
Recommended learning resources 6
Where to access unit resources 7
Key dates for this study period 7
Unit weekly activities 7

Welcome

This unit aims to provide students with detailed knowledge and understandings for analysis and design of power electronics circuits. The contents of this unit will cover key analytical methods for analysing power electronics circuits, and a wide range of power converter circuits in various industrial and renewable energy applications. Students will have the opportunity to design state-of-the-art power electronics converters such as buck and boost converters for different conduction modes and analyse the performance of different types of switches. The laboratory experiments will allow students to develop hands-on experience in implementing converter circuits which are essential components for any renewable energy systems. This unit lays foundations to the understanding and design of the power electronics components associated with renewable energy systems.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Samson Yu
Campus: Geelong Waurn Ponds
Email: samson.yu@deakin.edu.au
Phone: +61 3 522 73783
Other members of the team and how to contact them

There are no other staff involved in teaching this unit.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the unit have helped them to achieve the learning outcomes:

- An informative unit with useful theoretical and practical knowledge.
- Great practical experiments.
- Timely feedback and response.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who

have undertaken this unit in previous trimesters:

- Lab experiments have been refined with more detailed steps.
- Lecture recordings have been re-recorded.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Demonstrate a clear understanding of the characteristics of operation and application of power electronics circuits.	GLO1: Discipline-specific knowledge and capabilities
ULO2	Demonstrate the ability to solve analytical and design problems related to power electronics circuits and their applications.	GLO1: Discipline-specific knowledge and capabilities GLO4: Critical thinking GLO5: Problem solving
ULO3	Demonstrate the ability to critically analyse and communicate the performances of power electronics circuits and devices in terms of their efficiency and reliable operation.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO4: Critical thinking
ULO4	Demonstrate the ability to conduct experimental studies to understand and analyse the characteristics of power electronics devices and circuits. Also demonstrate the ability to design and implement power electronic circuits for practical applications.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy GLO7: Teamwork

Assessing your achievement of the unit learning outcomes

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	**Laboratory reports (group)**
Brief description of assessment task	Students will conduct three laboratory experiments to study and validate their theoretical understanding on the characteristics, operation and applications of power electronic devices and circuits. For each of the laboratory experimental studies, students will be required to work collegially in a team and submit written reports. These reports are group reports, which will reflect students' teamwork skills and technical capability for the unit content.

Detail of student output	This is a group assessment task. Students will be required to submit three written technical reports (10-page maximum each) that summarises the outcomes of each of the experimental works they carry out in the laboratory. Students will be assessed on: (i) Demonstration of their practical understanding on the characteristics and operation of power electronics circuits and devices. (ii) Demonstration of the ability to critically analyse the experimental results and relate them with the theoretical characteristics of power electronics circuits and devices. (iii) Ability to communicate experimental results through written report. Team submissions will be marked. There will be an individualised scaling factor applied based on peer and self-review to reflect your contribution and ability to work well within a team, moderated by the Unit Chair. If this self and peer review is not completed, a penalty of up to 25% of the available mark is deducted.
Grading and weighting (% total mark for unit)	45% (3 labs x 15% each) (Marked and Graded)
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1- Demonstrate a clear understanding of the characteristics of operation and application of power electronics circuits. ULO3 - Critically analyse and communicate the performances of power electronics circuits and devices in terms of their efficiency and reliable operation. ULO4 - Conduct experimental studies related to power electronics devices and circuits, individually and/or in a group, and communicate the experimental results through written reports.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - Through student ability to demonstrate both theoretical and practical understanding of characteristics and operational principles of power electronic devices and circuits. GLO2 - Through student ability to communicate the experimental results through written report. GLO3- Through the use of modern technologies in simulations and experiments and research in related fields, students will gain experience and improve their capability in digital literacy. GLO4 - Through student ability to critically analyse the experimental results and relate them with theoretical characteristics and operation of power electronic devices and circuits. GLO7 - Through demonstration of student ability to conduct experimental studies on power electronics devices and circuits in a team and communicate the results of the experimentation through a written group report.
How and when you will receive feedback on your work	Students will receive feedback on the reports during the trimester through the Unit site. Students are encouraged to seek additional feedback from the Unit Chair, if required.
When and how to submit your work	**Campus students:** Reports are due Sundays, 8pm AEST, two weeks after the laboratories (Weeks 5, 8 and 11). Online submission via the unit site. **Online students:** Online students will conduct all laboratory works during the intensive week. Three reports will be due in week 11 (Sunday 28 September 2025, 8pm AEST) via the unit site.

Summative assessment task 2

Title	**Problem-based Assignment Part A and Part B**

Brief description of assessment task	This assessment task requires students to individually respond to a set of descriptive, numerical, analytical and research problems related to characteristics, operational and design principles of power electronic devices and circuits for practical applications. Students will be assessed based on their understanding of the different power electronics circuits including converters and switches and relevant problem-solving skills to design and analyse such circuits.
Detail of student output	This is an individual task. Each task which will require written responses to a set of numerical and analytical questions.
Grading and weighting (% total mark for unit)	55% (Part A 25%, Part B 30%) (Marked and Graded)
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Demonstrate a clear understanding of the characteristics of operation and application of power electronics circuits. ULO2 - Solve analytical and design problems related to power electronic circuits and their applications. ULO4 - Demonstrate the ability to conduct experimental studies to understand and analyse the characteristics of power electronics devices and circuits. Also demonstrate the ability to design and implement power electronic circuits for practical applications.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - Through student ability to apply their understandings of the content to solve problems relevant to power electronic devices and circuits. GLO3 - Students will use modern media and technologies to find, use and communicate their research results, which helps them improve their digital literacy. GLO4 - Through student ability to apply their critical thinking skill to analyse mathematical and analytical problems. GLO5 - Through student ability to solve mathematical and analytical problems.
How and when you will receive feedback on your work	Students will receive feedback on problem-based assignment Part A during the trimester through the unit site within 2 weeks of their submissions, while the results for Part B will be withheld and factored into the final results of the unit. Students are encouraged to seek additional feedback from the Unit Chair, if required.
When and how to submit your work	**Part A** due in Week 7 (Sunday 24 August) **Part B** due in Week 12 (Sunday 5 October). Both assignments will need to be submitted online via the unit site by 8pm AEST. Further details will be provided on unit site at the start of the trimester.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 2 hour seminar per week, 3 x 2 hour practical experiences (laboratories) per trimester.

Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including a 1 x 2 hour seminar per week. Students must attend and participate in practical experiences (laboratories) at the Waurn Ponds (Geelong) Campus during the intensive activities as detailed in the unit site.

Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

For campus students the learning activities include watching recorded content and attending on-campus seminars and laboratories. For online students these activities include watching recorded content, attending online seminars and

participating in practical activities at the Geelong Waurn Ponds Campus at scheduled sessions during the trimester intensive week.

Students are also expected to spend time reading the practical resources provided and completing self-directed problem based exercises and assessment tasks.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SEE212 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

There is no prescribed textbook for this unit. Essential resources will be provided and include the unit site, Technical/Academic/Industry facilitators, laboratory facilities and the University Library.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

Recommended textbooks:

- **Fundamentals of Power Electronics** by, Robert W. Erickson and Dragan Maksimovic
  Publisher: Springer Science & Business Media, 3rd edn, 2020, ISBN: 9783030438814
- **Fundamentals of electric circuits** by Charles Alexander; Matthew N. O. Sadiku

Publisher: McGraw-Hill US Higher Education, 7th edn, 2021

Other suitable recommended readings for this unit are available in the University Library unit reading list.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 2 teaching period begins	Monday 7 July 2025
Census Date	Friday 15 August 2025
Intra-trimester break	Monday 11 August - Sunday 17 August 2025
Trimester 2 teaching period ends	Friday 26 September 2025
Study period (end-of-unit assessment/examination preparation period)	Monday 29 September - Friday 3 October 2025
End-of-unit assessment/examination period	Monday 6 October - Friday 17 October 2025
Inter-trimester break (the period between trimesters)	Monday 20 October - Friday 31 October 2025
Unit results released	Friday 31 October 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Special learning activities	Assessment due date
1	7 July 2025	Chapter 1 - Review of basic circuits, phasor, three- phase and complex power.		
2	14 July 2025	Chapter 1		
3	21 July 2025	Chapter 2 - Converter analysis principle		
4	28 July 2025	Chapter 2		
5^	4 August 2025	Chapter 2		**Assessment 1: Group Lab 1 report (for campus students)** due 10 August
6	18 August 2025	Chapter 3 - Equivalent circuit, loss and efficiency		
7	25 August 2025	Chapter 3 - Equivalent circuit, loss and efficiency; and Chapter 4 - Switches	**Intensive week for online students**	**Assessment 2: Problem-based assignment Part A** due 24 August
8	1 September 2025	Chapter 4 - Switches		**Assessment 1: Group Lab 2 report (for campus students)** due 7 September
9	8 September 2025	Chapter 4 - Switches; and Chapter 5 - Discontinuous Conduction Mode		
10	15 September 2025	Chapter 5		

Week	Commencing	Topic	Special learning activities	Assessment due date
11#	22 September 2025	Chapter 6 - Converter circuits; and Review		**Assessment 1: Group Lab 3 report (for campus students)** due 28 September **Assessment 1: Group Lab reports x 3 (for online students)** due 28 September
12	29 September 2025			Study period **Assessment 2: Problem-based assignment Part B** due 5 October
13-14	6 October 2025			End-of-unit assessment and exam period

^ Intra-trimester break: **Monday 11 August – Sunday 17 August 2025**
AFL Grand Final Eve public holiday: **Friday 26 September 2025** - University closed